

Consistency of Normal-Theory FIML under Non-normality and Missingness

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What is being proved

Two questions, kept separate because their answers differ.

1. **Non-normality.** Is the normal-theory (Gaussian) FIML estimator consistent for the mean and covariance when the data are non-normal? *Yes, unconditionally*, for complete data and under MCAR, assuming only finite second moments and a correctly specified structure. The argument is elementary and uses no distributional assumption at all.
2. **MAR.** Is it consistent under data missing at random? *Yes, but conditionally.* Consistency holds if and only if, for the incomplete variables, the conditional means given the always-observed causes are linear and the conditional covariances are homoscedastic. Marginal non-normality is harmless; conditional non-linearity or heteroscedasticity is not.

The second answer reconciles the textbook statement “FIML is consistent under MAR” (Rubin’s ignorability [4]; Arminger and Sobel [2]; Yuan and Bentler [7]) with the folklore that it is not, and with simulations in which the MAR bias looks like a small-sample effect that shrinks with N . Both camps are right about different regimes; the precise dividing line is Theorem 3 below.

Throughout, $X_1, \dots, X_N$ are i.i.d. p -vectors with finite second moments, true mean μ_0 and covariance $\Sigma_0 \succ 0$. The saturated parameter is $\eta = (\mu, \text{vech } \Sigma)$ with true value η_0 . A structural model writes $\eta = \eta(\theta)$. “Consistent” means convergence in probability to the full-data target as $N \rightarrow \infty$.

1 The saturated Gaussian MLE is the sample moments

The whole non-normality question collapses once we notice that with complete data the saturated Gaussian likelihood has a closed-form maximizer that never saw a normal assumption.

Lemma 1. *With complete data, the maximizer of the Gaussian log-likelihood $\sum_i \left[-\frac{1}{2} \log |\Sigma| - \frac{1}{2} (x_i - \mu)' \Sigma^{-1} (x_i - \mu) \right]$ over $(\mu, \Sigma \succ 0)$ is*

$$\hat{\mu} = \bar{x} = \frac{1}{N} \sum_i x_i, \quad \hat{\Sigma} = S = \frac{1}{N} \sum_i (x_i - \bar{x})(x_i - \bar{x})'.$$

Proof. The mean score $\sum_i \Sigma^{-1} (x_i - \mu) = 0$ gives $\hat{\mu} = \bar{x}$ for any $\Sigma \succ 0$. Substituting and maximizing over Σ is the standard Gaussian profile problem whose solution is the empirical second-moment matrix S . $\square$

So the saturated FIML point estimate *is* the method-of-moments estimator of (μ, Σ) . It is a Gaussian *quasi*-likelihood when the data are non-normal (the general misspecified-likelihood theory of White [6]), but its location is fixed by the first two empirical moments alone.

Theorem 1 (Complete-data consistency, distribution-free). *If $\mathbb{E}\|X\|^2 < \infty$ then $\hat{\eta} \xrightarrow{p} \eta_0$. If in addition the structure is correctly specified, with $\eta(\theta_0) = \eta_0$ for a unique identified θ_0 , and the discrepancy $\theta \mapsto F\{\hat{\eta}; \eta(\theta)\}$ is continuous with a unique minimizer, then $\hat{\theta} \xrightarrow{p} \theta_0$.*

Proof. By the law of large numbers $\bar{x} \xrightarrow{p} \mu_0$ and $S \xrightarrow{p} \Sigma_0$, so $\hat{\eta} \xrightarrow{p} \eta_0$ by Lemma 1. The fitting function F is minimized at the truth and is continuous in its first argument, so the standard argmin (Wald) consistency theorem, together with the continuous mapping theorem, gives $\hat{\theta} \xrightarrow{p} \theta_0$. Normality is used nowhere. $\square$

Remark. This is consistency of the *point estimate*. Validity of normal-theory standard errors and of the unscaled likelihood-ratio reference law is a strictly stronger property that does require either asymptotic-robustness conditions [3, 1] or a sandwich correction [6]. Theorem 1 is exactly why the point estimates survive non-normality while the test statistics need the FMG/MLR machinery of the companion note.

2 MCAR: still distribution-free

With missing data the saturated MLE loses the closed form of Lemma 1 (it needs EM), so we work directly with the FIML score and show it is an *unbiased estimating equation* at the full-data moments. This makes the h_i argument fully explicit.

The FIML likelihood and its score. Let unit i have observed index set $o_i \subseteq \{1, \dots, p\}$ and observed subvector x_{i,o_i} . FIML contributes the Gaussian *marginal* log-density of that subvector,

$$\ell_i(\eta) = -\frac{1}{2} \log \det \Sigma_{o_i o_i} - \frac{1}{2} (x_{i,o_i} - \mu_{o_i})' \Sigma_{o_i o_i}^{-1} (x_{i,o_i} - \mu_{o_i}),$$

dropping the constant $-\frac{1}{2}|o_i| \log 2\pi$. Write the residual $r_i = x_{i,o_i} - \mu_{o_i}$ and the observed-block precision $W_i = \Sigma_{o_i o_i}^{-1}$. Differentiating in the saturated coordinates $\eta = (\mu, \text{vech } \Sigma)$ gives the casewise score $h_i(\eta) = \partial \ell_i / \partial \eta$, with a mean block and a covariance block,

$$\frac{\partial \ell_i}{\partial \mu_{o_i}} = W_i r_i, \quad \frac{\partial \ell_i}{\partial \Sigma_{o_i o_i}} = \frac{1}{2} (W_i r_i r_i' W_i - W_i),$$

and *zero* in every coordinate indexing an unobserved mean μ_j ($j \notin o_i$) or a covariance entry that touches an unobserved variable. (The covariance block is read in the symmetric matrix and mapped to vech by the duplication matrix, which has full column rank and so does not move the root.) Thus h_i is a *sparse* vector in η -space, supported exactly on the moments pattern o_i observes; FIML is the stacked estimating equation $\sum_i h_i(\eta) = 0$, in which each moment's row aggregates only the units that see it.

Unbiasedness at the truth. Read off the two blocks: the mean block has mean zero and the covariance block has mean zero exactly when

$$\mathbb{E}[r_i] = 0 \quad \text{and} \quad \mathbb{E}[r_i r_i'] = \Sigma_{0,o_i o_i},$$

i.e. when the observed subvector carries its *population* first two moments. (For the covariance block, $\mathbb{E}[W_i r_i r_i' W_i - W_i] = W_i (\mathbb{E}[r_i r_i'] - \Sigma_{0,o_i o_i}) W_i = 0$ at η_0 .) That is the only distributional fact the argument needs, and it is exactly what MCAR supplies.

Theorem 2 (MCAR consistency, distribution-free). *Suppose missingness is MCAR, the response pattern R independent of X , with finite second moments, and every entry of (μ_0, Σ_0) identified by some observed pattern. Then $\hat{\eta} \xrightarrow{P} \eta_0$; with a correctly specified structure, $\hat{\theta} \xrightarrow{P} \theta_0$.*

Proof. Condition on the realized pattern. Under MCAR, $R \perp X$, so conditioning on $R_i = o$ does not change the law of $X_{i,o}$: it keeps the population marginal, whence $\mathbb{E}[r_i \mid R_i = o] = 0$ and $\mathbb{E}[r_i r'_i \mid R_i = o] = \Sigma_{0,oo}$ at η_0 . By the display above both score blocks then have conditional expectation zero, and the unobserved coordinates of h_i are identically zero, so $\mathbb{E}\{h_i(\eta_0) \mid R_i = o\} = 0$ for every pattern o . Averaging over the pattern distribution gives $\mathbb{E}\{h_i(\eta_0)\} = 0$: the full-data η_0 is the population root of the FIML estimating equation. Under the stated identification (each moment seen by some pattern) the root is unique, so the Z -estimator $\hat{\eta}$ satisfies $\hat{\eta} \xrightarrow{p} \eta_0$; the argmin step of Theorem 1 carries it to $\hat{\theta}$. $\square$

The single load-bearing step is $\mathbb{E}[r_i \mid R_i = o] = 0$: under MCAR, selecting a pattern does not tilt the observed-block mean. That is exactly the step that breaks under MAR, where selection looks at the observed values and shifts their conditional mean; Section 3 is the accounting of that shift.

3 MAR: the real question

Now let the missingness depend on the data. We use the structure of the experiment-23 (Savalei–Bentler) design, which is also the canonical MAR setup: a block A of variables is *always observed* and is the only thing the mechanism may look at,

$$P(R = r \mid X) = g_r(X_A),$$

and the remaining variables B are observed in various patterns selected by $g_r(X_A)$. Write the always-observed/conditional decomposition of the Gaussian model.

Lemma 2 (Variation-independent factorization). *The Gaussian density factors as*

$$\phi(x; \mu, \Sigma) = \phi(x_A; \mu_A, \Sigma_{AA}) \phi(x_B; \alpha + \Gamma x_A, \Sigma_{BB \cdot A}),$$

with $\Gamma = \Sigma_{BA} \Sigma_{AA}^{-1}$, $\alpha = \mu_B - \Gamma \mu_A$, and $\Sigma_{BB \cdot A} = \Sigma_{BB} - \Sigma_{BA} \Sigma_{AA}^{-1} \Sigma_{AB}$. The map $(\mu, \Sigma) \leftrightarrow (\mu_A, \Sigma_{AA}, \alpha, \Gamma, \Sigma_{BB \cdot A})$ is a smooth bijection with the two coordinate blocks varying independently. Consequently the FIML log-likelihood splits exactly,

$$\underbrace{\sum_i \log \phi(x_{i,A}; \mu_A, \Sigma_{AA})}_{\text{all units: } A \text{ always observed}} + \underbrace{\sum_{i: B \text{ obs.}} \log \phi(x_{i,B}; \alpha + \Gamma x_{i,A}, \Sigma_{BB \cdot A})}_{\text{selected units: Gaussian regression of } B \text{ on } A},$$

and the two halves are maximized separately.

The first half is a marginal model on the fully observed A , so by Lemma 1 it estimates $(\mu_{A,0}, \Sigma_{AA,0})$ consistently, distribution-free. Everything hinges on the second half: a Gaussian linear-homoscedastic regression of B on A , fit on the *selected* complete cases.

Theorem 3 (MAR consistency condition). *Under MAR-on-A with finite second moments and FIML identification, the Gaussian FIML estimator is consistent for (μ_0, Σ_0) if and only if the full-data conditional moments of the incomplete block satisfy*

(i) $\mathbb{E}[X_B | X_A] = \alpha_0 + \Gamma_0 X_A$ is **linear** in X_A , and

(ii) $\text{Cov}[X_B | X_A] = \Sigma_{BB \cdot A,0}$ does **not depend** on X_A (homoscedastic).

Under (i)–(ii) consistency holds for arbitrarily non-normal marginals. If either fails, $\hat{\eta}$ carries an asymptotic bias whose magnitude is set by the selection reweighting of X_A .

Proof. The A -block is consistent by Lemma 2. For the conditional block, the complete cases are selected with probability $g(X_A)$ depending *only* on X_A . Hence the conditional law $X_B | X_A$ is unchanged by selection ($f_{\text{sel}}(x_B | x_A) = f(x_B | x_A)$); only the marginal of X_A is reweighted to $f_{\text{sel}}(x_A) \propto g(x_A)f(x_A)$.

Mean and slope. The Gaussian conditional MLE solves the selected normal equations $\mathbb{E}_{\text{sel}}[(1, X_A')'(X_B - \alpha - \Gamma X_A)] = 0$. Evaluate the left side at the true full-data coefficients (α_0, Γ_0) and use the tower property over $X_B | X_A$ (selection-invariant):

$$\mathbb{E}_{\text{sel}}[(1, X_A')' r(X_A)], \quad r(X_A) = \mathbb{E}[X_B | X_A] - \alpha_0 - \Gamma_0 X_A.$$

Under (i), $r \equiv 0$, so this vanishes for *every* selection weight g ; (α_0, Γ_0) is therefore the population root of the selected equation and is estimated consistently.

Conditional covariance. The residual MLE targets $\mathbb{E}_{\text{sel}}[(X_B - \alpha_0 - \Gamma_0 X_A)(\cdot)'] = \mathbb{E}_{\text{sel}}[\text{Cov}(X_B | X_A)] + \mathbb{E}_{\text{sel}}[r r']$. Under (i) the second term is zero, and under (ii) the first equals the constant $\Sigma_{BB \cdot A,0}$ regardless of the g -weighting. So $\Sigma_{BB \cdot A,0}$ is recovered.

Mapping back through the continuous bijection of Lemma 2 ($\mu_B = \alpha + \Gamma \mu_A$, $\Sigma_{BA} = \Gamma \Sigma_{AA}$, $\Sigma_{BB} = \Sigma_{BB \cdot A} + \Gamma \Sigma_{AA} \Gamma'$) yields consistency for (μ_0, Σ_0) .

Necessity. If (i) fails, $r \not\equiv 0$, and the selected normal equations are solved by the linear projection over $f_{\text{sel}}(x_A) \propto g(x_A)f(x_A)$, which differs from the

full-population projection whenever g is non-constant: $\mathbb{E}_{\text{sel}}[(1, X'_A)'r] \neq 0$ generically, so the probability limit of $(\hat{\alpha}, \hat{\Gamma})$ is biased. If (ii) fails, $\mathbb{E}_{\text{sel}}[\text{Cov}(X_B | X_A)]$ is a g -weighted average of a non-constant conditional covariance and differs from the unweighted $\Sigma_{BB \cdot A, 0}$. Either way $\text{plim } \hat{\eta} \neq \eta_0$. $\square$

Corollary 1 (Why marginal shape is irrelevant). *Skewness and kurtosis of the marginal law of X enter only through the shape of $f(x_B | x_A)$ and $f(x_A)$, never through conditions (i)–(ii), which constrain only the first two conditional moments. A heavily skewed or heavy-tailed marginal with linear, homoscedastic conditionals (the regime that Vale–Maurelli, NORTA, and most copula constructions approximate well) therefore gives a **consistent** FIML estimator under MAR.*

Remark (The elliptical sharpening). Elliptical non-normal laws are the cleanest cautionary example. They have $\mathbb{E}[X_B | X_A]$ *linear* (so (i) holds) but $\text{Cov}[X_B | X_A] = w(X_A) \Sigma_{BB \cdot A}$ *heteroscedastic* (so (ii) fails unless w is constant). Under MAR on an elliptical population, FIML stays consistent for μ_0 and for the slope/regression part $\Sigma_{BA} \Sigma_{AA}^{-1}$, but is biased for the residual covariance $\Sigma_{BB \cdot A}$, hence for Σ_{BB} . The mean structure is robust; the dependent-block variance is the fragile piece. “FIML is consistent under MAR” must be qualified at least this far.

4 Reconciliation, and what the simulations show

The literature claim is correct under its hypotheses. Rubin’s ignorability [4] delivers valid likelihood inference under MAR *when the complete-data model is correctly specified*. For the Gaussian model, “correctly specified” does not mean full normality; it means the first two conditional moments are Gaussian-shaped, which is exactly (i)–(ii). That is why the claim survives marginal non-normality. Yuan and Bentler [7] and Arminger and Sobel [2] prove consistency and asymptotic normality of the normal-theory estimator under MAR; their robustness is to *marginal* non-normality, which enters only the sandwich meat J_η (the third- and fourth-moment information) and so moves the standard errors and the test reference law, *not* the location of the estimating-equation root. Conditional misspecification, by contrast, moves the root itself; that is the irreducible “estimator-target” bias the companion note flags as something no reference-law correction (MLR, SB, FMG) can repair [5].

So the textbook statement and the folklore are not in conflict. Read “correct model” as “correct first two conditional moments” and both are true:

consistent under (i)–(ii), inconsistent otherwise.

Reading the simulations. Two phenomena look alike at finite N and must be separated:

- (a) *Genuine asymptotic bias* (conditions (i)–(ii) fail): the bias converges to a nonzero constant as $N \rightarrow \infty$.
- (b) *Finite-sample bias of a consistent estimator* (conditions hold): the bias is $O(N^{-1})$ or $O(N^{-1/2})$ and vanishes.

The experiment-23 MAR is mild on exactly the axis that matters here: selection is a coarse *sign* rule on the always-observed x_1, x_2 , and the Vale–Maurelli / inverse-Gaussian / power-law generators keep the missing-given-observed conditional means close to linear and the conditional covariances close to constant. The design therefore nearly satisfies (i)–(ii), its true asymptotic bias is ≈ 0 , and what appears in the tables is case (b): a small-sample bias that shrinks with N . That is fully consistent with the estimator being consistent.

How to settle it empirically. To decide (a) vs (b) for any given cell, regress the absolute parameter bias on $1/N$ (or $1/\sqrt{N}$) across the sample-size ladder: a near-zero intercept certifies consistency (the slope is the benign finite-sample term); a clearly positive intercept certifies inconsistency. The test-statistic analogue is the note’s $\widehat{\text{n\acute{c}p}} = \text{mean_base} - \text{mean_trace}$, which must shrink toward zero with N under (b) and persist under (a). Predicted outcome for experiment 23: intercepts indistinguishable from zero, i.e. a finite-sample artifact, not inconsistency.

Verdict.

- Non-normal complete data and MCAR: consistent, unconditionally (Theorems 1 and 2).
- MAR: consistent if and only if the incomplete-on-observed conditional means are linear and conditional covariances homoscedastic (Theorem 3); marginal non-normality alone does not break it (Corollary 1). Your instinct is right for *general* non-normal MAR; the paper’s claim is right under the conditional-moment conditions, which the experiment-23 generators very nearly meet, so the residual MAR bias you see is finite-sample and should vanish with N .

References

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